

ASSIGNMENT 2

Guidelines

Marks 100

VIVA

Submission	Date
Final Submission (Online via Moodle)	24/05/2026
VIVA (On campus) based on the submissions (the schedule will be made available via Moodle)	After Final exam(mention later)

Problem Description

Task 1

This assignment introduces a **web-based Fuel QR Registration System** for a fuel distribution network. The system automates the manual process of registering vehicles and managing fuel quotas. Its friendly web UI allows fuel station staff and administrators to handle registrations in a convenient, efficient, and simplified manner. The system has a **REST API** to manage resources in the backend.

Task 2

Provide a step-by-step guide on updating the code to **add a new service**. You will be asked to do the same in your demonstrations.

REST API (Fuel QR Registration)

- Register a Vehicle
- Show all Registered Vehicles
- Find a Vehicle by Registration Number (RegNo)
- Find Vehicles by Owner's First Name
- Find Vehicles by Owner's Last Name
- Find Vehicles by Owner's Email
- Find Vehicles by Nearest Fuel Station
- Find Vehicles by Fuel Type (Petrol/Diesel/Electric)
- Find Vehicles by Owner's NIC
- Update Vehicle by Registration Number
- Update Vehicle by Owner's First Name
- Delete Vehicle by Registration Number

Vehicle Record Format (JSON Array)

```
{  
  "RegNo": "ABC-1234",  
  "FirstName": "Kamal",  
  "LastName": "Perera",  
  "Email": "kamal@gmail.com",  
  "NearestStation": "Colombo City Fuel",  
  "FuelType": "Petrol",  
  "OwnerNIC": "123456789V",  
  "VehicleModel": "Toyota Vitz",  
  "QRCode": "ABC1234QR"  
}
```

Assignment Requirements

Create a Website to fulfill the following requirements:

- Implement **both Client-side and Server-side** integration
- **Front End:** HTML + Bootstrap
- **jQuery:** 5 different uses in the site
- **Ajax:** For data fetch from the server
- **JSON:** For communication
- **Back End:** REST API with Node.js
- **Database:** MongoDB